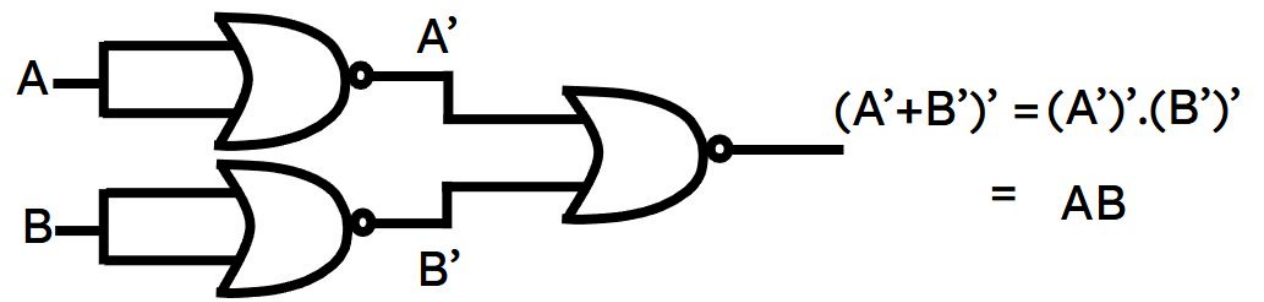
Lecture 3

Logic Gates & Boolean Algebra



CSE260 : Digital Logic Design

Binary Logic

- Binary logic consists of binary variables and logical operations.
- Variables are designated by letters such as A, B, C, x, y, z etc. with only 2 possible values: 1 and 0.
- Logic operations: and, or, not etc.

Logic Gates

- The most basic digital devices are called gates.
- A gate has one or more inputs and produces an output that is a function of the current input values.
- The relationship between the input and the output is based on a certain logic.

Truth Table

- Provides a listing of every possible combination of inputs and its corresponding outputs.

INPUTS	OUTPUTS
. . .	. . .
. . .	. . .

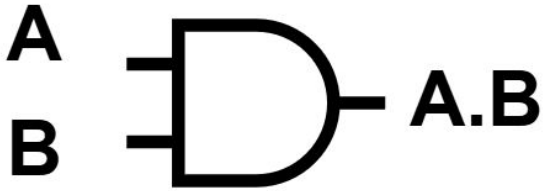
All Logic Gates

NOT	Basic Gates
AND	
OR	
NAND	Universal Gates
NOR	
XOR	Exclusive Gates
XNOR	

2-input AND gate

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

1 = on
0 = off

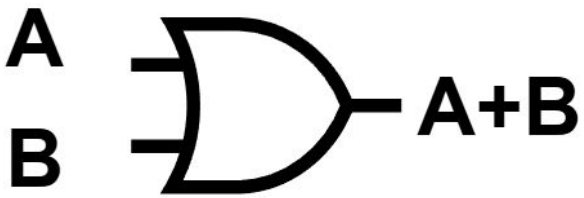


Output will be 1 only when all inputs are 1

2-input OR gate

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

1 = on
0 = off

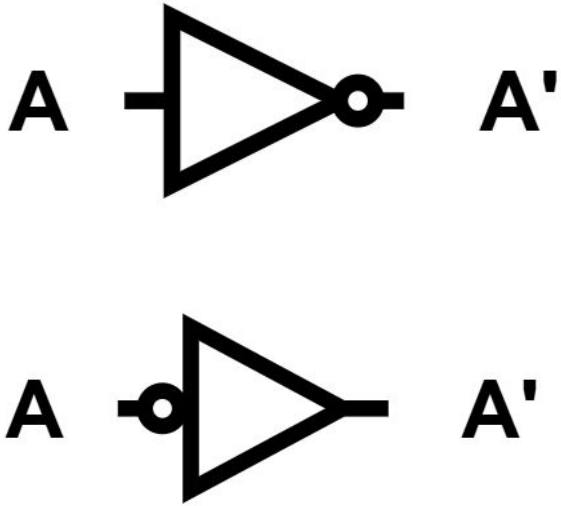


Output will be 1 when at least one input is 1

NOT gate (Inverter)

A	A’
0	1
1	0

1 = on
0 = off



Output will be the inverse of the input

Universal & Exclusive Gate

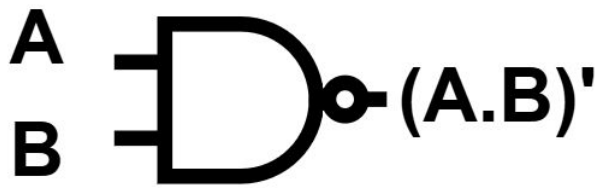
- NAND and NOR are also known as universal gates
- A **universal gate** is a **gate** which can implement any other **gate**

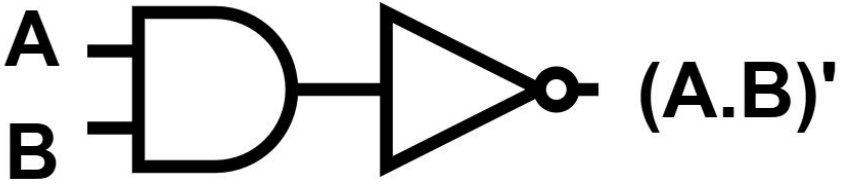
- XOR and XNOR are known as Exclusive Gates
- An **exclusive gate** is a **gate** that outputs based on exclusive input conditions

2-input NAND gate

A	B	(A.B)’
0	0	1
0	1	1
1	0	1
1	1	0

1 = on
0 = off



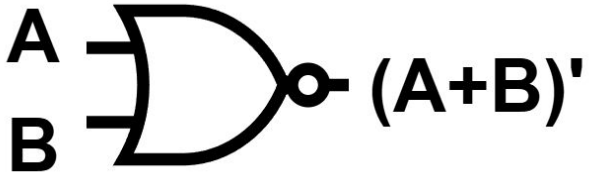


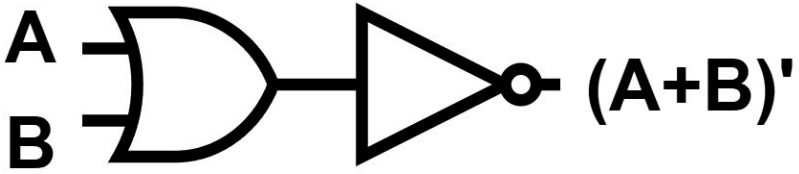
Inverse of AND.
Output will be 1 when at least one input is 0

2-input NOR gate

A	B	(A+B)’
0	0	1
0	1	0
1	0	0
1	1	0

1 = on
0 = off

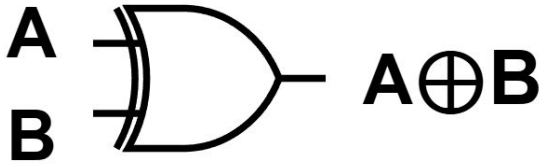




Inverse of OR
Output will be 1 when all input is 0

2-input XOR gate

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0



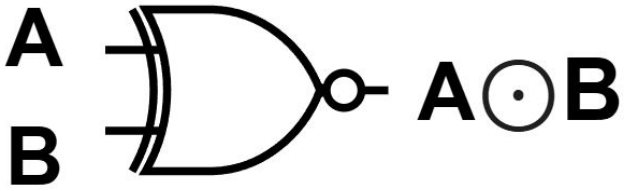
1 = on
0 = off

Output will be 1 for odd number of 1s in input

Trivia: Do you know that A⊕B and **A'B+AB'** are the same thing? Create the truth table to verify.

2-input XNOR gate

A	B	A⊙B
0	0	1
0	1	0
1	0	0
1	1	1



1 = on
0 = off

Inverse of XOR

Trivia: Do you know that A⊙B and AB+A'B' are the same thing? Create the truth table to verify.

Proof using Truth Table

- Prove that: x . (y + z) = (x . y) + (x . z)

(i) Construct truth table for LHS & RHS of above equality.

x	y	z	y+z	x.(y+z)	x.y	x.z	(x.y)+(x.z)
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

(ii) Check that LHS = RHS
Postulate is SATISFIED because output column 5 & 8 (for LHS & RHS expressions) are equal for all cases.

BOOLEAN ALGEBRA

Boolean Algebra

- Like any other deductive mathematical system, defined with a set of elements, a set of operators and a number of axioms or postulates.

- In Boolean algebra, set consists at least 2 variables say x & y, with 2 binary operations {+} and {.} and 1 unary operation {‘}

Basic Theorems of Boolean Algebra

- Theorems can be proved using the truth table method. (Exercise: Prove De-Morgan's theorem using the truth table.)
- They can also be proved by algebraic manipulation using axioms/postulates or other basic theorems.

Postulates and Theorems of Boolean Algebra

Postulate 2: Identity	(a) x + 0 = x	(b) x . 1 = x
Postulate 3: Commutative	(a) x + y = y + x	(b) x . y = y . x
Postulate 4: Distributive	(a) x(y+z) = xy + xz	(b) x + yz = (x+y).(x+z)
Postulate 5: Complement	(a) x + x’ = 1	(b) x . x’ = 0
Theorem 1:	(a) x + x = x	(b) x . x = x
Theorem 2:	(a) x + 1 = 1	(b) x . 0 = 0
Theorem 3: Involution	(x’)’ = x	
Theorem 4: Associative	(a) x+(y+z) = (x+y) +z	(b) x.(y.z) = (x.y).z
Theorem 5: DeMorgan	(a) (x + y)’ = x’ . y’	(b) (xy)’ = x’ + y’
Theorem 6: Absorption	(a) x + xy = x	(b) x. (x + y) = x

All are very very important!

Basic Theorems of Boolean Algebra

- Theorem 2a can be proved by:

$$x + 1 = x+(x+x') \quad \text{(complement)}$$
$$= (x+x)+x' \quad \text{(Th. 4)}$$
$$= x+x' \quad \text{(complement)}$$
$$= 1$$

By duality, theorem 2b:

$$x.(0) = 0$$

Note: There can be other ways of making this proof. See Morris Mano

Basic Theorems of Boolean Algebra

- Theorem 6a (absorption) can be proved by:

$$
\begin{aligned}
x + x.y &= x.1 + x.y && \text{(identity)} \\
&= x.(1 + y) && \text{(distributivity)} \\
&= x.(y + 1) && \text{(commutativity)} \\
&= x.1 && \text{(Theorem 2a)} \\
&= x && \text{(identity)}
\end{aligned}
$$

By duality, theorem 6b:

$$x.(x+y) = x$$

Try prove this by algebraic manipulation.

Duality

- **Duality Principle** – every valid Boolean expression (equality) remains valid if the operators and identity elements are interchanged, as follows:

$$+ \leftrightarrow .$$
$$1 \leftrightarrow 0$$

Example: Given the expression

$a + (b.c) = (a+b).(a+c)$

then its dual expression is

$a \ . (b+c) = (a.b) + (a.c)$

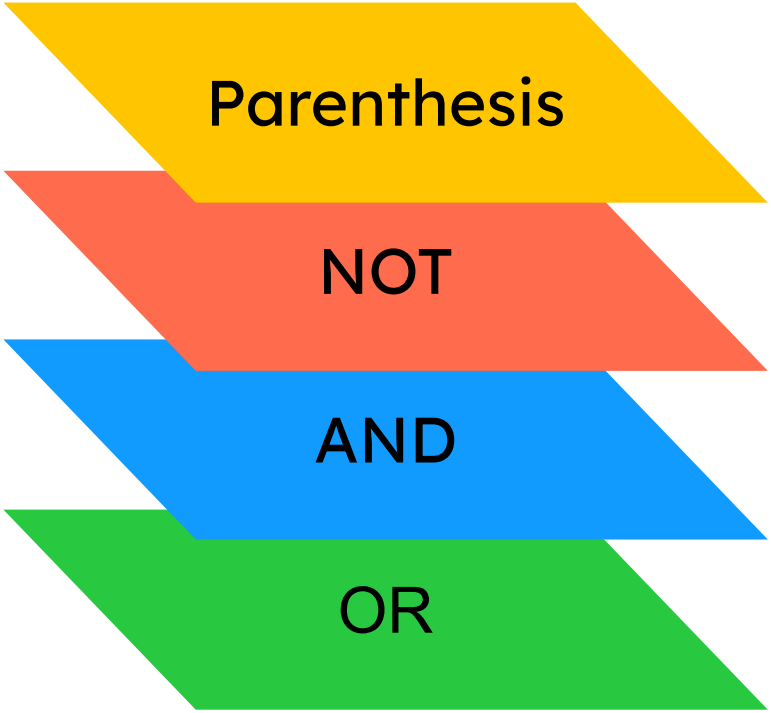
Duality

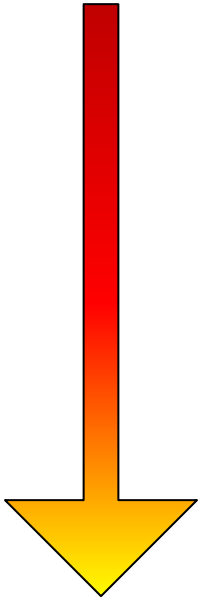
- Duality gives free theorems –
  “two for the price of one”.
  You prove one theorem and the other comes for free!

- If $(x+y+z)' = x'.y.'z'$ is valid, then its dual is also valid:
  $(x.y.z)' = x'+y'+z$’

- If $x + 1 = 1$ is valid, then its dual is also valid:
  $x \,.\, 0 = 0$

Operator Precedence





Highest

Lowest

Simplification

- We can simply a boolean function in the following ways

Using Boolean Algebra

K-Map (Lecture 5)

Tabulation (Self-Study)

Boolean Functions (Solve ?)

Example:

$F1 = xyz'$

$F2 = x + y'z$

$F3 = (x'y'z) + (x'yz) + (xy')$

$F4 = xy' + x'z$

x	y	z	F1	F2	F3	F4
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	0

From the truth table, F3=F4.
Can you also prove by algebraic manipulation that F3=F4?

Boolean Functions (Solve ?)

If F3=(x′y′z)+(x′yz)+(xy′) and F4=xy′+x′z, prove F3=F4 by algebraic manipulation.

Solution:

F3=
(x′y′z)+(x′yz)+(xy′)
= x’y’z+x’yz+xy’
=x’z(y’+y)+xy’
=x’z(1)+xy’
=x’z+xy’
=F4

Postulate 2: Identity	(a) x + 0 = x	(b) x . 1 = x
Postulate 3: Commutative	(a) x + y = y + x	(b) x . y = y . x
Postulate 4: Distributive	(a) x(y+z) = xy + xz	(b) x + yz = (x+y).(x+z)
Postulate 5: Complement	(a) x + x’ = 1	(b) x . x’ = 0
Theorem 1:	(a) x + x = x	(b) x . x = x
Theorem 2:	(a) x + 1 = 1	(b) x . 0 = 0
Theorem 3: Involution	(x’)’ = x	
Theorem 4: Associative	(a) x+(y+z) = (x+y) +z	(b) x.(y.z) = (x.y).z
Theorem 5: DeMorgan	(a) (x + y)’ = x’ . y’	(b) (xy)’ = x’ + y’
Theorem 6: Absorption	(a) x + xy = x	(b) x. (x + y) = x
Theorem 7: Consensus	(a) AB + BC +A'C = AB + A'C	

Try it yourself

a)Simplify to minimum literals: xy+xy'
b)Reduce to 6 literals: BC+AC'+AB+BCD
Simplify the following equations
c) x+x'y
d) x(x'+y)
e) x'y'z+x'yz+xy'

Postulate 2: Identity	(a) x + 0 = x	(b) x . 1 = x
Postulate 3: Commutative	(a) x + y = y + x	(b) x . y = y . x
Postulate 4: Distributive	(a) x(y+z) = xy + xz	(b) x + yz = (x+y).(x+z)
Postulate 5: Complement	(a) x + x' = 1	(b) x . x' = 0
Theorem 1:	(a) x + x = x	(b) x . x = x
Theorem 2:	(a) x + 1 = 1	(b) x . 0 = 0
Theorem 3: Involution	(x')' = x	
Theorem 4: Associative	(a) x+(y+z) = (x+y) +z	(b) x.(y.z) = (x.y).z
Theorem 5: DeMorgan	(a) (x + y)' = x' . y'	(b) (xy)' = x' + y'
Theorem 6: Absorption	(a) x + xy = x	(b) x. (x + y) = x
Theorem 7: Consensus	(a) AB + BC +A'C = AB + A'C	

Try it yourself

Solution:

a) xy+xy’ =x(y+y’) =x(1) =x

b)BC+AC’+AB+BCD

=BC(1+D)+AC’+AB

=BC(1)+AC’+AB

=BC+AB+AC’

=B(C+A)+AC’

c) x+x’y =(x+x’).(x+y) =1.(x+y) =x+y

d) x(x’+y) =xx’+xy =0+xy =xy

e) x’y’z+x’yz+xy’ =x’z(y’+y)+xy’ =x’z+xy’

Now Try Proving Using Truth Table!!!

Complementing a function

1. Take dual of the function
2. Complement each literals

Example: F1= x'yz'+x'y'z

1. Dual of the function F1 is

   (x'+y+z').(x'+y'+z)

2. Complement each literal=

   (x+y'+z)(x+y+z')

Therefore, F1'= (x+y'+z)(x+y+z')

Same as applying De-Morgan's law on the function

Try it yourself

a)What is the complement of
F2=x(y'z'+yz)

Solution:

Step 1: Duality: x+(y'+z')(y+z)
Step 2: Complement= x'+(y+z)(y'+z')
Therefore, F2' = x'+(y+z)(y'+z')

More Practice

Simplify the following Boolean expression to a minimum number literals:

- a) $xy + xy'$
- b) $(x + y)(x + y')$
- c) $xyz + x'y + xyz'$
- d) $(A+B)'(A'+B')'$
- e) $(AB)'(A' + B)(B' + B)$
- f) $(A + C)(AD + AD') + AC + C$

More Practice [Solution]

a) $xy + xy' = x (y+y')= x.1 = x$

b) $(x+y)(x+y') = xx + xy' + yx+yy' = x + xy' + xy + 0 = x (1+ y' + y) = x.1= x$

Also $(x+y)(x+y') =x+ yy' = x + 0 = x$

c) $xyz + x'y + xyz' = xy(z+z') + x'y = xy +x'y = y(x+x')=y$

d) $(A+B)'(A'+B')'= (A'B').(AB) = 0$

e) A' [Find the process yourself]

f) A + C [Find the process yourself]

Practice! Practice! Practice!

Find the complement of the following expressions:

- a) $xy'+x'y$
- b) $(AB'+C)D'+E$
- c) $(x+y'+z)(x'+z')(x+y)$

Practice! Practice! Practice!

Solution:

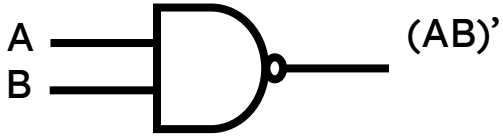
a) xy′+x′y = (x+y′) . (x′+y) = (x′+y).(x+y′)

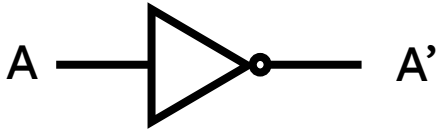
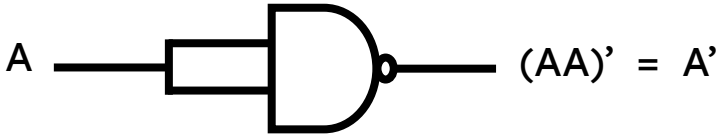
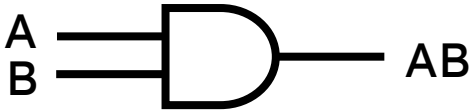
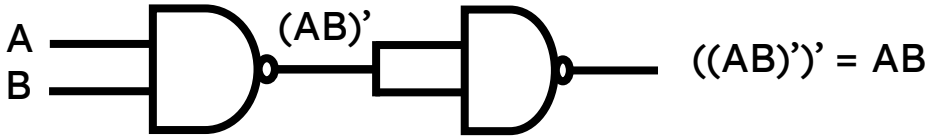
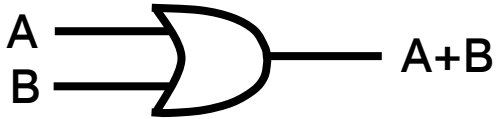
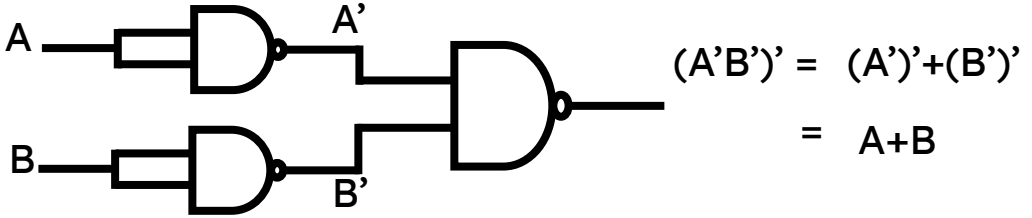
b) (AB′+C)D′+E = (A+B’.C)+D’.E = (A’+B.C’)+D.E’

c) (x+y′+z)(x′+z′)(x+y) = (x.y′.z)+(x′.z′)+(x.y) = (x’.y.z’)+(x.z)+(x’.y’)

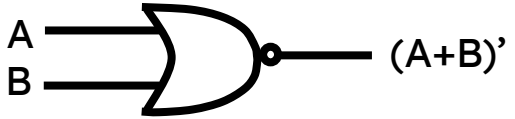
Using NAND and NOR to Build Other Gates and Functions

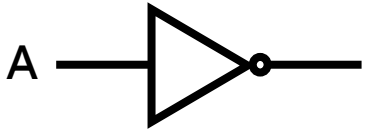
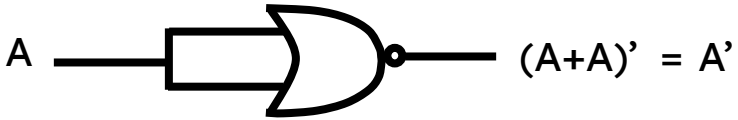
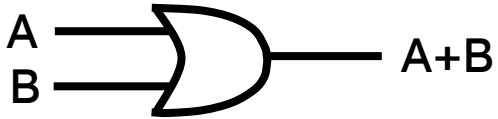
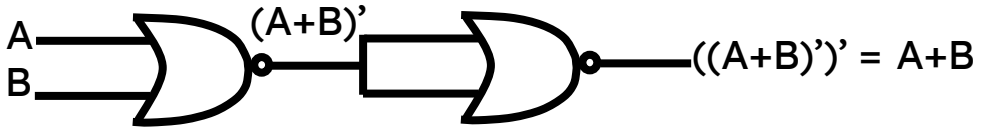
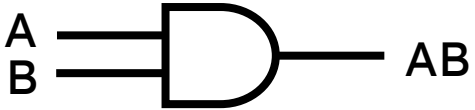
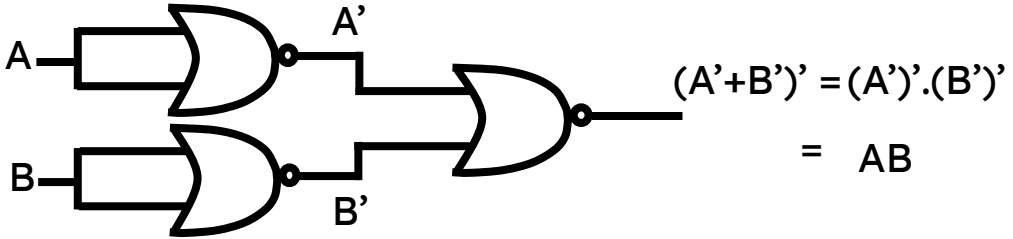
Basic gates using NAND gate



NOT		
AND		
OR		

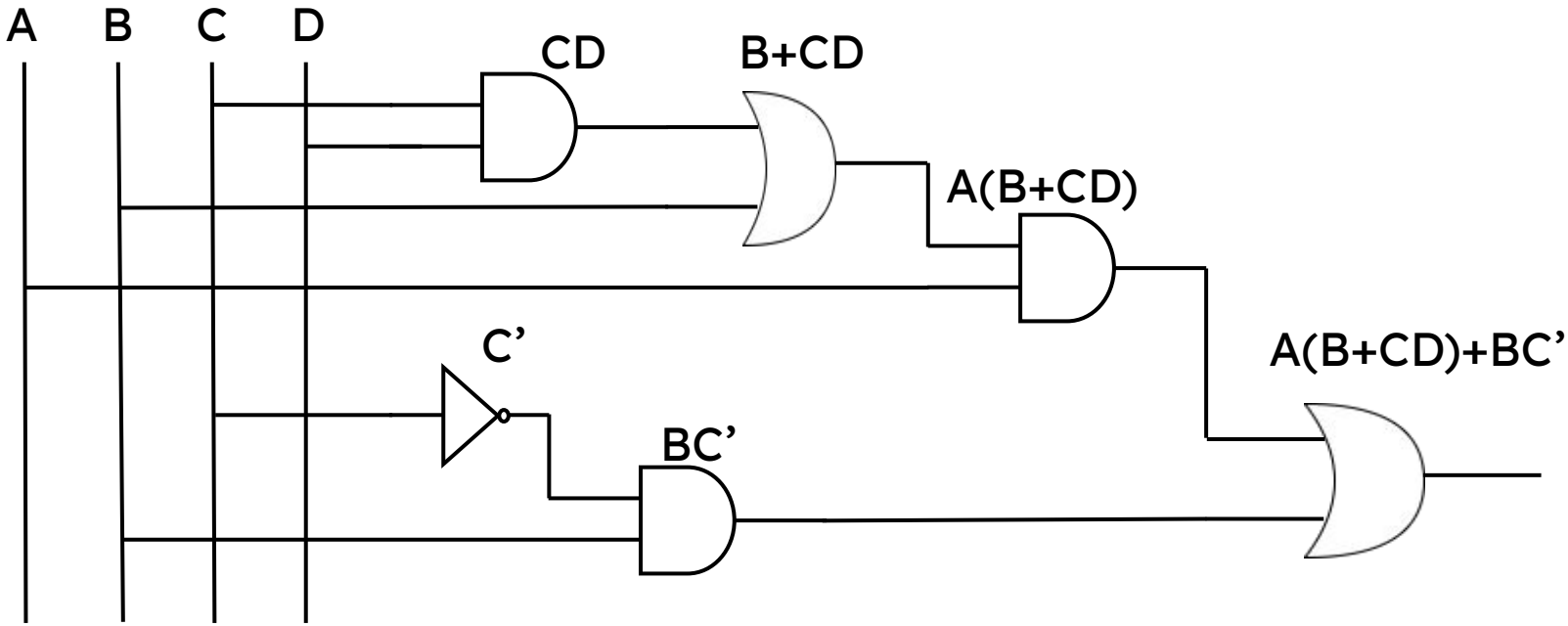
Basic gates using NOR gate



NOT		
OR		
AND		

Exercise

Use AND, OR, NOT to represent F=A(B+CD)+BC’



Use NAND to Implement A Function

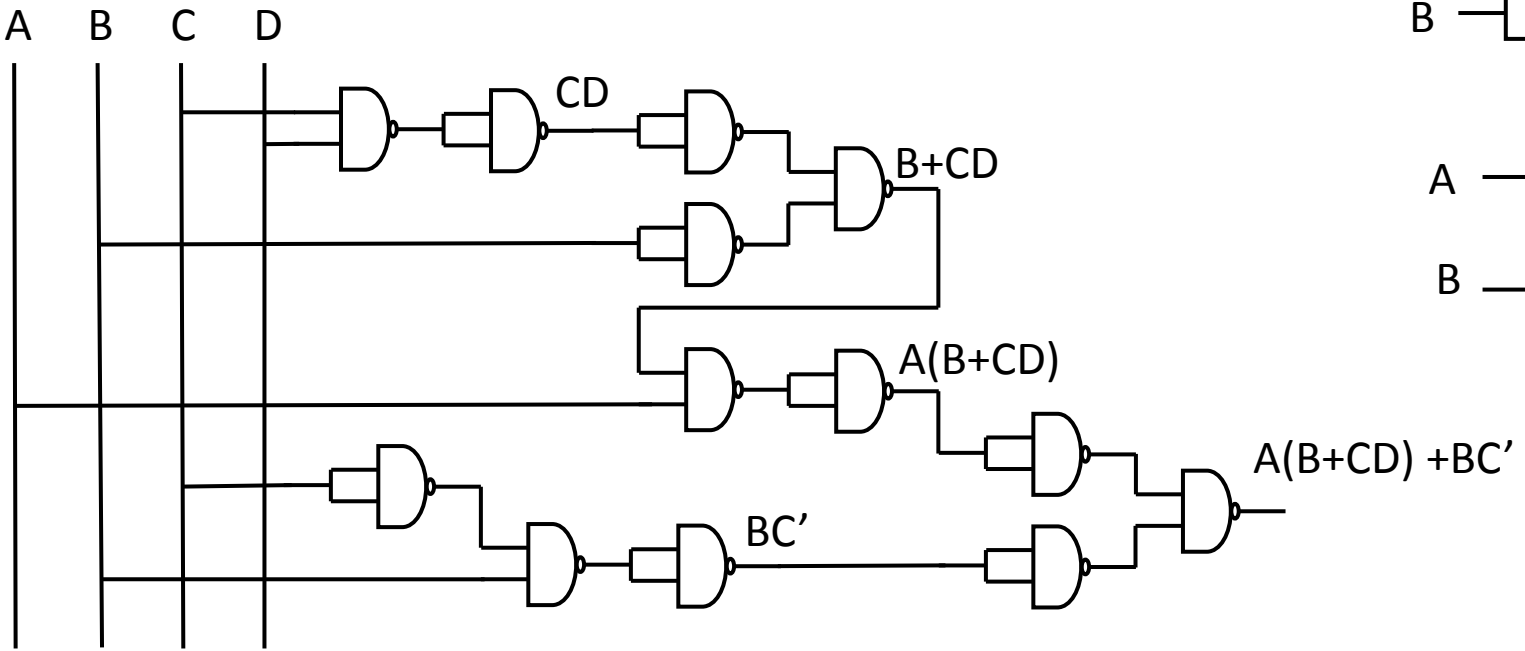
Steps:

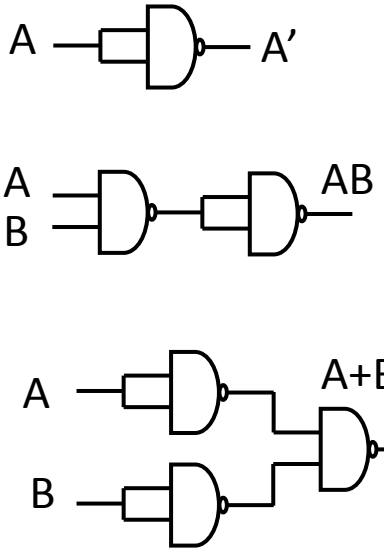
1. If an expression is given, represent it using AND,OR and NOT gates
2. Draw each gate with equivalent NAND representation
3. Remove any 2 cascading inverters
4. Remove inverters from single input connection and replace input with its complement.

Tutorial

Exercise

Use NAND to represent F=A(B+CD)+BC’





Use NOR to Implement A Function

Steps:

1. If an expression is given, represent it using AND,OR and NOT gates
2. Draw each gate with equivalent NOR representation
3. Remove any 2 cascading inverters
4. Remove inverters from single input connection and replace input with its complement.

Tutorial

Exercise

Use NOR to represent F=A(B+CD)+BC’

